Medical image analysis challenge

Final project

February 4, 2026

1 Introduction

As part of the *Deep learning for medical image analysis* course, you are expected to perform a group project that accounts for **30%** of your grade. For this project, you will work on a real-world medical imaging problem in which you can demonstrate what you have learned during the course. We invite you to participate in a so-called *challenge*. This is a benchmark or competition in which many different methods are evaluated on the same data set. Challenges allow an objective and fair comparison between different methods on the same public data set [4]. This is important especially in medical imaging, because data sharing can be difficult due to privacy concerns.

For this project, you should team up with **three** other students, i.e., each team consists of four members. Because this is an interdisciplinary project, it pays off to team up with fellow students who have complementary skills. For example, if you have a strong medical background but lack experience in Python programming, we recommend that you work with a computer science student. Conversely, a computer science student could learn from a fellow student with a medical background. Working in teams like this reflects the interdisciplinary nature of medical image analysis. When you **sign yourself up** in one of the groups on Canvas, please keep this in mind.

Each year, many different challenges are organized. Many of these are held in conjunction with large scientific conferences such as MICCAI and ISBI. The Grand Challenge website organized by RadboudUMC aims to collect both the released data as well as the developed algorithms for all of these challenges. The aim of this project is that you use the knowledge and skills gained in the course to tackle one of these challenges. We have pre-selected three popular and widely used challenges that have previously been organized. We ask you to choose **one** of these as a goal for your project. If you have good reasons to want to participate in a challenge that is not listed here (e.g., because it matches a planned internship or PhD project), please discuss this beforehand with us.

2 Challenges

We have pre-selected three challenges that are widely used for the development and evaluation of new medical image analysis algorithms. The difficulty of the challenges varies; we indicate this with stars. For all these challenges,

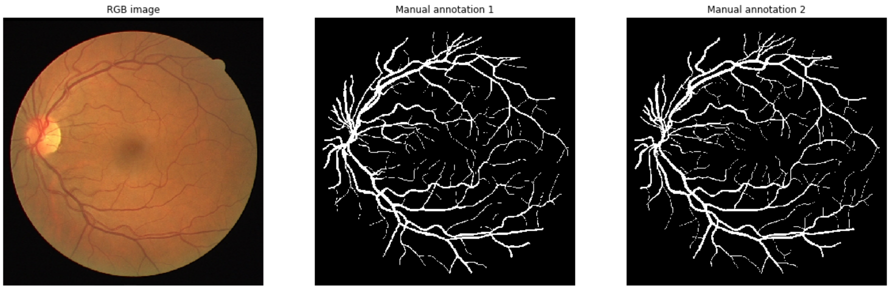

Figure 1: Example of an RGB image provided in DRIVE and two manual annotation masks for the blood vessels by human observers.

training data and *test* data with annotations are available. This means that you can develop and optimize your method using one dataset, and evaluate its performance on the unseen test dataset. All these challenges use an online evaluation platform, which you can use to assess your developed algorithm.

In addition to the training and test set provided by the challenge, we have selected an additional *secret test set* for each challenge that you will use to test the generalization performance of your method.

2.1 DRIVE ★★☆☆☆

Description from website: *The Digital Retinal Images for Vessel Extraction (DRIVE) database was established to enable comparative studies on segmentation of blood vessels in retinal images [5]. Retinal vessel segmentation and delineation of morphological attributes of retinal blood vessels, such as length, width, tortuosity, branching patterns, and angles are utilized for the diagnosis, screening, treatment, and evaluation of various cardiovascular and ophthalmologic diseases such as diabetes, hypertension, arteriosclerosis, and choroidal neovascularization. Automatic detection and analysis of the vasculature can assist in the implementation of screening programs for diabetic retinopathy and can aid research on the relationship between vessel tortuosity and hypertensive retinopathy, vessel diameter measurement in relation to diagnosis of hypertension, and computer-assisted laser surgery. Automatic generation of retinal maps and extraction of branch points have been used for temporal or multimodal image registration and retinal image mosaic synthesis. Moreover, the retinal vascular tree is found to be unique for each individual and can be used for biometric identification.*

The database contains 40 2D retinal fundus images (Fig. 1) with manually annotated vessel masks. Your goal is to segment images as accurately as possible using a deep learning algorithm. Performance is evaluated using the Dice similarity metric. This is an overlap metric that ranges from 0 (poor segmentation) to 1 (excellent segmentation). You can use the `grand-challenge.org` framework to evaluate your results (once a day) and you will end up in a public leaderboard that compares your results to those of others.

Challenge This data set has been around for almost twenty years and has thus been heavily used. This means that the bar to actually beat the current

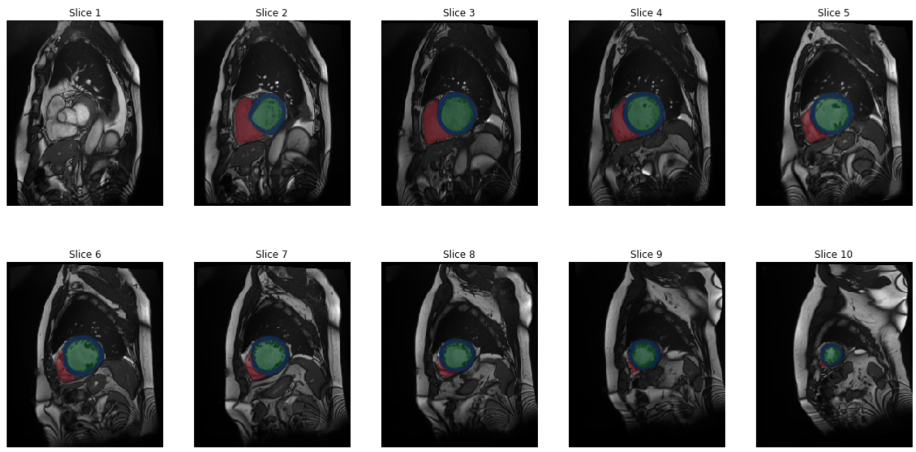

Figure 2: Example of an ACDC cardiac cine MR image with segmentation mask. Heart structures are indicated in green (left ventricle), blue (myocardium), and red (right ventricle). These images are taken at ten different short-axis cross-sections along the left ventricle's main axis.

state-of-the-art will be very high. You are *challenged* to think out of the box and come up with new ideas on how to solve this problem. Moreover, because this is a **relatively** simple task (compared to ACDC and ASOCA), you are encouraged to also investigate, e.g., how your method generalizes to other retinal fundus image datasets like STARE.

2.2 ACDC ★★★★☆

Description from website: *The goal of this contest is two-fold. First to compare the performance of automatic methods on the segmentation of the left ventricular endocardium and epicardium as the right ventricular endocardium for both end diastolic and end systolic phase instances. Second, to compare the performance of automatic methods for the classification of the examinations in five classes (normal case, heart failure with infarction, dilated cardiomyopathy, hypertrophic cardiomyopathy, abnormal right ventricle). The overall ACDC dataset was created from real clinical exams acquired at the University Hospital of Dijon. Acquired data were fully anonymized and handled within the regulations set by the local ethical committee of the Hospital of Dijon (France). Our dataset covers several well-defined pathologies with enough cases to (1) properly train machine learning methods and (2) clearly assess the variations of the main physiological parameters obtained from cine-MRI (in particular diastolic volume and ejection fraction). The dataset is composed of 150 exams (all from different patients) divided into 5 evenly distributed subgroups (4 pathological plus 1 healthy subject groups) as described below. Furthermore, each patient comes with the following additional information: weight, height, as well as the diastolic and systolic phase instants.*

As the text says, the Automatic Cardiac Diagnosis Challenge (ACDC) consists of two parts [1]. First, the multi-class segmentation of multi-slice 2D + time (so basically 4D) cardiac cine MR images (Fig. 2). This is a challenging

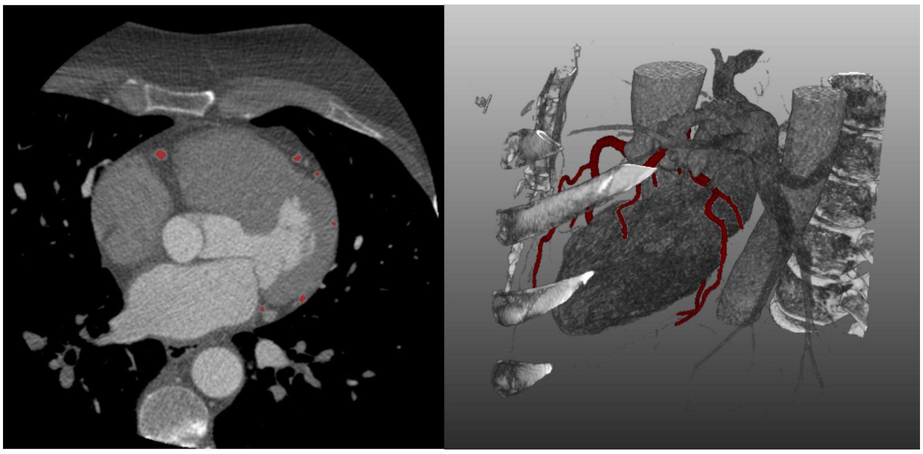

Figure 3: Example CT images in the ASOCA challenge, with coronary arteries shown in red. *left* 2D axial slice, *right* 3D rendering of the CT scan of the same patient, which shows the position of the heart between the ribs and the spine.

task, which many people have already worked on [2]. If you do this well, it can be extremely useful for, e.g., the detection of heart failure. The second part of the challenge is a classification task. Can you, based on just these images and some basic patient dimensions (height, weight), determine which class a patient belongs to? Is the patient healthy, or does he/she have one of four common cardiac diseases? The **focus** of your project should be on the segmentation task, but you are free also to give the classification task a try.

You can download all data from this website. The set is split into training images (100 patients) and test images (50 patients). You can use the training data to develop your data, but in the end, you should report the results on the test set. This means that you should obtain segmentation masks for 100 test images (two per patient) and quantitatively evaluate results in these patients. To evaluate your segmentations, you can use code provided by the challenge organizers.

Challenge There is quite some variability in the data set. Images have different sizes, different intensity distributions, and segmentation is much harder in some parts of the heart than in others.

2.3 ASOCA ★★★★★

The Automatic Segmentation of Coronary Arteries (ASOCA) challenge focuses on the segmentation of coronary arteries. You will do this in 3D cardiac CT volumes that have been acquired with ECG-triggering: the image is acquired at a particular point in the cardiac cycle in which the heart is moving as little as possible so that you can clearly see the coronary arteries. These are the small arteries that supply your heart muscle with oxygenated blood. The challenge organizers provide 60 CT images: 30 of normal/healthy patients and 30 of patients with cardiovascular disease. The goal is to classify each voxel in the image as either belonging to the coronary artery class or the background class (Fig. 3). Typically, the coronary arteries of patients with cardiovascular disease are more difficult to segment. You might find a complete description of the data set in a recently published paper [3].

Challenge You are looking for relatively small structures in a large image. Arteries are typically just a few voxels wide, whereas the full image can easily contain 10 million voxels. This is a highly imbalanced segmentation problem. This challenge runs on `grand-challenge.org` and uses a live leaderboard. This way, you will immediately know how well your method performs compared to others. The challenge provides 40 training images *with* reference masks and 20 test images *without* reference masks. You can use the training data to develop your method, but in the end, you should report results on the test images, and for this, you should use the online platform. This means that you should register for the challenge on `https://asoca.grand-challenge.org/`. While you wait for your registration to be approved, you can already download the full data set here.

3 Plan of approach

Your first step will be to form groups. You can do this on Canvas by signing up for one of the groups that we have already set up. You then have to decide with your group on which challenge you want to work. Please take a look at some example data that we have put on SURFdrive. After you have selected the challenge that you want to work on, we expect you to write a short plan of approach. This will not be graded, but allows us to provide you with some feedback on your plans. This plan of approach should be max. 1 A4 and should include the following information:

- The names and study programmes of your team members
- The name of the challenge that you will work on
- A figure that you have created yourself, visualizing an image from the dataset that you will be using
- A coarse description of the method that you intend to use
- At least two references to relevant scientific papers (other than the ones in this document)
- A planning, in which you describe for each week what your aims are

This information should be handed in via Canvas no later than **Monday February 16, 5pm**. Note that only one plan should be submitted per team/group, but it's **important** that you provide the names of all teammates.

4 Project execution

You are expected to work on the project during the course. We count on your independence and initiative to meet up with your teammates. The tutorials will also provide you with ample code examples for your project. In principle, the computing solutions for the tutorial (UT JupyterLab, Google Colab, or your own machine) should be sufficient, but if you run into an issue with this, please let us know, and we might find an alternative. There will be one dedicated session where you can work on your project code with our supervision and ask for feedback (**March 12**). You will have to register yourself on the challenge website to download the data and evaluate your results. If you have any questions regarding the process or if you get stuck when working with the data, you can ask those on Canvas. You can, of course, also ask questions when you see us at the tutorials. Each team will be assigned one *mentor* that you can reach out to if you run into problems in your project. We will let you know who your mentor is.

5 Products

Your group will receive a grade based on three products: a presentation of your work, a (short) report, and the availability of source code.

5.1 Project presentation

At the end of the course (**Tuesday, April 7**), we will organize a scientific symposium in which you will present your results. Each team gets 3 minutes to *pitch* their work (you can use slides for this), followed by a poster session in which you present a poster about your method and results. This symposium will be open to everyone in the course, but also to other students and staff members. There will be a **prize** for the best presentation and a prize for the best-performing method on each of the challenges.

5.2 Report

Your main result is, of course, a working method, but also a report in which you describe what you have done. Please adhere to the Springer Lecture Notes in Computer Science (LNCS) paper format to write your report. You can find the template on Overleaf, please stick to **8 pages max. with up to 2 pages for references**. You are allowed to submit up to ten pages, excluding supplementary materials. Your report should include (at least)

- **Abstract** 150-250 words in which you describe your project. Writing a good abstract is not easy, but there is plenty of advice that you can find on the internet.
- **Introduction** A description of the problem, the challenge, and what your main contribution is. It might help to follow a structured approach here, such as the SCQA method.
- **Materials and methods** Describe the data that you have used and how you are solving the problem. This includes information such as the neural network architecture that you use, but also data augmentation strategies and evaluation criteria. Also include a link to your code (see Sec. 5.2.1).
- **Experiments and results** List the experiments that you have performed and which results you have obtained. Don't forget to add some figures (it

is an imaging course after all). Experiments can take on many forms. For example, you can evaluate how different hyperparameter settings affect the performance of your method, you can investigate the interpretability of your method (**Lecture 7**), or you can assess performance on different subsets of your data.

- **Discussion** What is your main finding? How does your work relate to other methods working on the same data? Remember to add references.

5.2.1 Code

Reproducibility is very important in science (and for companies). We ask you to upload your code to GitLab or to GitHub and add a link to your code in the report.

6 Important dates

All deadlines are 5 pm. Products should be handed in via Canvas Assignments.

- **Thursday, February 5** Start forming teams of four students.
- **Monday, February 16** Submit your plan of approach (Sec. 3) on Canvas.
- **Tuesday, March 12** Q&A session projects.
- **Tuesday, April 7** Project presentation.
- **Thursday, April 23** Hand in final report.

References

[1] Olivier Bernard, Alain Lalande, Clement Zotti, Frederick Cervenansky, Xin Yang, Pheng-Ann Heng, Irem Cetin, Karim Lekadir, Oscar Camara, Miguel Angel Gonzalez Ballester, et al. Deep learning techniques for automatic MRI cardiac multi-structures segmentation and diagnosis: is the problem solved? *IEEE transactions on medical imaging*, 37(11):2514–2525, 2018.

[2] Francesco Galati, Sébastien Ourselin, and Maria A Zuluaga. From accuracy to reliability and robustness in cardiac magnetic resonance image segmentation: A review. *Applied Sciences*, 12(8):3936, 2022.

[3] R Gharleghi, D Adikari, K Ellenberger, M Webster, C Ellis, A Sowmya, S Ooi, and S Beier. Annotated computed tomography coronary angiogram images and associated data of normal and diseased arteries. *Scientific Data*, 10(1):128, 2023.

[4] Lena Maier-Hein, Matthias Eisenmann, Annika Reinke, Sinan Onogur, Marko Stankovic, Patrick Scholz, Tal Arbel, Hrvoje Bogunovic, Andrew P Bradley, Aaron Carass, et al. Why rankings of biomedical image analysis competitions should be interpreted with care. *Nature communications*, 9(1):1–13, 2018.

[5] Joes Staal, Michael D Abràmoff, Meindert Niemeijer, Max A Viergever, and Bram Van Ginneken. Ridge-based vessel segmentation in color images of the retina. *IEEE transactions on medical imaging*, 23(4):501–509, 2004.